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Tectonic evaluation of the Indochina Block during Jurassic-Cretaceous from palaeomagnetic results of Mesozoic redbeds in central and southern Lao PDR

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ABSTRACT

Rock magnetic and palaeomagnetic studies were performed on Mesozoic redbeds collected from the central and southern Laos, the northeastern and the eastern parts of the Khorat Plateau on the Indochina Block. Totally 606 samples from 56 sites were sampled and standard palaeomagnetic experiments were made on them. Positive fold tests are demonstrated for redbeds of Lower and Upper Cretaceous, while insignificant fold test is resulted for Lower Jurassic redbeds. The remanence carrying minerals defined from thermomagnetic measurement, AF and Thermal demagnetizations and back-field IRM measurements are both magnetite and hematite. The positive fold test argues that the remanent magnetization of magnetite or titanomagnetite and hematite in the redbeds is the primary and occurred before folding. The mean palaeomagnetic poles for Lower Jurassic, Lower Cretaceous, and Upper Cretaceous are defined at Plat./Plon. = 56.0°N/178.5°E (A_{95} = 2.6°), 63. 3°N/170.2°E (A_{95} = 6.9°), and 67.0°N/180.8°E (A_{95} = 4.9°), respectively. Our palaeomagnetic results indicate a latitudinal translations (clockwise rotations) of the Indochina Block with respect to the South China Block of −10.8 ± 8.8° (16.4 ± 9.0°); −11.1 ± 6.2° (17.8 ± 6.8°); and −5.3 ± 4.7° (13.3 ± 5.0°), for Lower Jurassic, Lower Cretaceous, and Upper Cretaceous, respectively. These results indicate a latitudinal movement of the Indochina Block of about 5–11° (translation of about 750–1700 km in the southeastward direction along the Red River Fault) and clockwise rotation of 13–18° with respect to the South China Block. The estimated palaeoposition of the Khorat Plateau at ca. 21–26°N during Jurassic to Cretaceous argues for a close relation to the Sichuan Basin in the southwest of South China Block. These results confirm that the central part of the Indochina Block has acted like a rigid plate since Jurassic time and the results also support an earlier extrusion model for Indochina.

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1. Introduction

Intercontinental collisions and deformations have taken place in a variety of modes in the Asia continent. The deformations of Southeast Asia have been modeled in a number of laboratory and numerical experiments during the last two decades (e.g. Tapponnier et al., 1982; Cohen and Morgan, 1986; Houseman and England, 1993; Royden et al., 2008). An extrusion tectonic model has been proposed to explain the tectonic of Southeast Asia after the Pre-Tertiary India-Asia collision which occurred in the Lower Eocene (ca. 50 Ma) (e.g. Molnar and Tapponnier, 1975; England and Houseman, 1986; Beck et al., 1995; Rowley, 1996; Tong et al., 2008) or in the Oligocene (ca. 35 Ma) (Aitchison et al., 2007). The southeastward movement of Indochina along the Ailao Shan-Red River (ASRR) shear zone is partly supported by geological evidences (Leloup et al., 1995, 2001; Wang et al., 1998, 2000, 2001; Gilley et al., 2003; Replumaz and Tapponnier, 2003; Replumaz et al., 2004; Ali et al., 2010) and geochronological data (Lacassin et al., 1997; Gilley et al., 2003). Palaeomagnetic studies have been performed in South China and Indochina blocks in attempts to reconstruct tectonic events in these regions (Zhu et al., 1988; Kent et al., 1986; Gilder et al., 1993, 1999; Morinaga and Liu, 2004; Li et al., 2005; Bai et al., 1998; Yokoyama et al., 2001; Enkin et al., 1992; Tanaka et al., 2008; Takemoto et al., 2009; Otofuji et al., 2010, 2012; Cung and Geissman, 2013). On the basis

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of reliable palaeomagnetic data from the Indochina Block a consecutive translation and rotation of Indochina during Mesozoic has been suggested (Yang, 1992; Yang and Besse, 1993; Bhongsuwan and Elming, 2000; Charusiri et al., 2006; Takemoto et al., 2005, 2009; Tanaka et al., 2008; Otofuji et al., 2010, 2012; Cung and Geissman, 2013). However, such translation and rotation interpreted from paleomagnetic data seems to have varied within the block, i.e. no significant rotation has been indicated for the parts, northwest and south of Vietnam when comparing with the Eurasia Apparent Polar Wander Path (APWP) (Takemoto et al., 2005; Otofuji et al., 2012; Cung and Geissman, 2013), while palaeomagnetic results from the Khorat Plateau Basin in the central part of the Indochina Block indicate a clockwise rotation and latitudinal translation relative to the South China Block (Yang and Besse, 1993; Bhongsuwan and Elming, 2000; Charusiri et al., 2006; Takemoto et al., 2009). These differences may be the results of complex internal deformation of the Indochina Block and parts of the Sundaland caused by the Indian–Asia collision. In order to test the coherency of the Indochina Block and to constrain the number of tectonic models, palaeomagnetic data from other parts of the blocks are necessary.

In this paper, we are going to provide new palaeomagnetic results and propose a tectonic model for the Indochina Block. This model is based on interpretation of palaeomagnetic data from Jurassic to Cretaceous redbeds in Laos. Anisotropy of magnetic susceptibility (AMS) data is also provided in order to determine eventual deformation of the studied rocks that may have affected the remanence directions.

2. Geological setting

The Indochina Peninsula includes the Indochina Block (INC) and part of the Shan-Thai Block (ST). The ST comprises eastern Myanmar, western Thailand, western peninsular Malaysia, and northern Sumatra (Fig. 1). The INC comprising northeastern and eastern Thailand, Laos, Cambodia, and parts of Vietnam is separated from the South China Block (SCB) by the NW–SE trending active Ailao Shan-Red River Fault (ASRRF) system, and is bounded by the Three Pagodas (TPF) and Wang Chao faults (WCF) in the southwest (Morley, 2001; Morley et al., 2001; Kornsawan and Morley, 2002; Akciz et al., 2008). The boundary between the Shan-Thai and the Indochina blocks is expressed by the Nan-Uttaradit Suture (NUS) in the west (Barr and MacDonald, 1987, 1991; Hada et al., 1999; Metcalfe, 1999) (Fig. 1a). There are several redbed basins of Mesozoic age within the Indochina Peninsula (BGMRY, 1990; Tien et al., 1991; Mantajit et al., 2002). The Khorat Plateau Basin, located in the northeastern Thailand and extending to western Laos and northern Cambodia, is the largest basin in the central part of the INC. The basin mostly comprises upper rock units of the INC, which is dominated by Mesozoic continental redbed sequences from the principal rock unit of the Khorat Group (Mantajit et al., 2002). The Khorat Group mainly comprises continental red and grey sandstone, siltstone, and shale. The Mesozoic sedimentary rocks in the central and southern Laos are also dominated by rocks of the Khorat Group. The age of the Khorat Group ranges from Upper Triassic to Cretaceous (Workman, 1977; Chonglakmani and Sattayarak, 1978; Buffetaut and Ingavat, 1987; DMR, 1999) or Upper Jurassic to Lower Cretaceous (Racey et al., 1997; Racey, 2009; Racey and Goodall, 2009; Buffetaut et al., 2006, 2009). The ages of the rocks in this study are obtained from the 1:1,000,000 geological and mineral resources map of Lao PDR (Department of Geology and Mines, 1991) and the study is focused on Lower Jurassic to Upper Cretaceous red beds exposed in the northeastern and eastern part of Khorat Plateau (in the central and southern Laos). These strata are similar to those of the Khorat Group with a large non-marine depositional system in the mainland Southeast Asia.

3. Sampling and laboratory procedures

Mesozoic redbeds were sampled from 56 sites (total of 606 samples) in central and southern Laos (Fig. 1b) using a portable gasoline-powered core drill. The orientation of the core samples was determined using both sun and magnetic compasses. The location of sampling sites was positioned using a GPS.

For laboratory procedure, 2–3 standard specimens of 2.5 cm in diameter and 2.2 cm in length were cut from each core sample. The remanent magnetizations were measured using a spinner magnetometer JR-6 (Agico, Czech Republic) and a DC-SQUID magnetometer (2G-Enterprises, USA) in the palaeomagnetic laboratories of the Prince of Songkla University, Thailand, Luleå University of Technology, and Lund University, Sweden. To evaluate magnetic stability and eliminate viscous magnetic components, progressive stepwise alternating field demagnetization up to 100 mT was performed with coils incorporated in the DC-SQUID system. Progressive stepwise thermal demagnetization (12–15 steps, up to 700 °C) was performed using MMTD18 (Magnetic Measurement, England) for high coercivity samples. Demagnetization results were analyzed using orthogonal plots (Zijderveld, 1967) and the components of remanent magnetizations were defined using principal component analysis (Kirschvink, 1980). The Super-IAPD software (Torsvik et al., 2000) was used for the componential analysis. At least four demagnetization steps were used to define remanence vectors. The site mean and formation mean directions were calculated using the statistics of Fisher (Fisher, 1953). Direction-correction (DC) fold test for individual sample mean directions at the 95% confidence level was performed using the method proposed by Enkin (2003). The test was run on Pmagtools 4.2a software written by Mark Hounslow (Hounslow, 2006). The magnitude of magnetic susceptibility and the orientations of the principal axes of Anisotropy of Magnetic Susceptibility (AMS) were measured at room temperature, using a low field Kappabridge KLY-3S (Agico, Czech Republic) in order to define the magnetic fabric of the rocks. The technique of Jelinek (1978) was used for statistical analyses of the AMS data. The AMS parameters, including shape parameter (T) and corrected anisotropy degree (Pj) proposed by Jelinek (1981), were used to define magnetic fabric and degree of anisotropy of the rocks, respectively. For analyses of magnetic minerals a CS3 furnace (Agico, Czech Republic) installed with the Kappabridge KLY-3S was used to measure the temperature dependence of magnetic susceptibility.

Saturation of isothermal remanent magnetization (SIRM) using a field of 1 T and the back-field IRM performed on a reversed field generated by an electromagnet were also performed step by step up to 500 mT. The IRM was measured using a Minispin magnetometer (Molspin, England).

4. Results and discussions

4.1. Anisotropy of Magnetic Susceptibility (AMS)

The mean bulk susceptibility of the redbeds varies between 13.7×10^{-6} and 542.5×10^{-6} SI, suggesting a low concentration of ferromagnetic and/or paramagnetic minerals in the rocks (Tarling and Hrouda, 1993) and the degree of anisotropy (Pj) is less than 8% ($Pj < 1.08$; Fig. 2). On the basis of the low degree of anisotropy the rock samples are interpreted to be undeformed or only weakly deformed (e.g. Tarling and Hrouda, 1993; Parés et al., 1999; Frizon De Lamotte et al., 2002; Parés, 2004; Borradaile and Jackson, 2010). This also suggests that the remanence directions have not been affected by any tectonic deformation. The shape parameter (T) of rocks in most sites is positive ($T > 0$) indicating predominantly oblate magnetic fabric most probably plate-like hematite. The majority of sites are characterized by well grouped

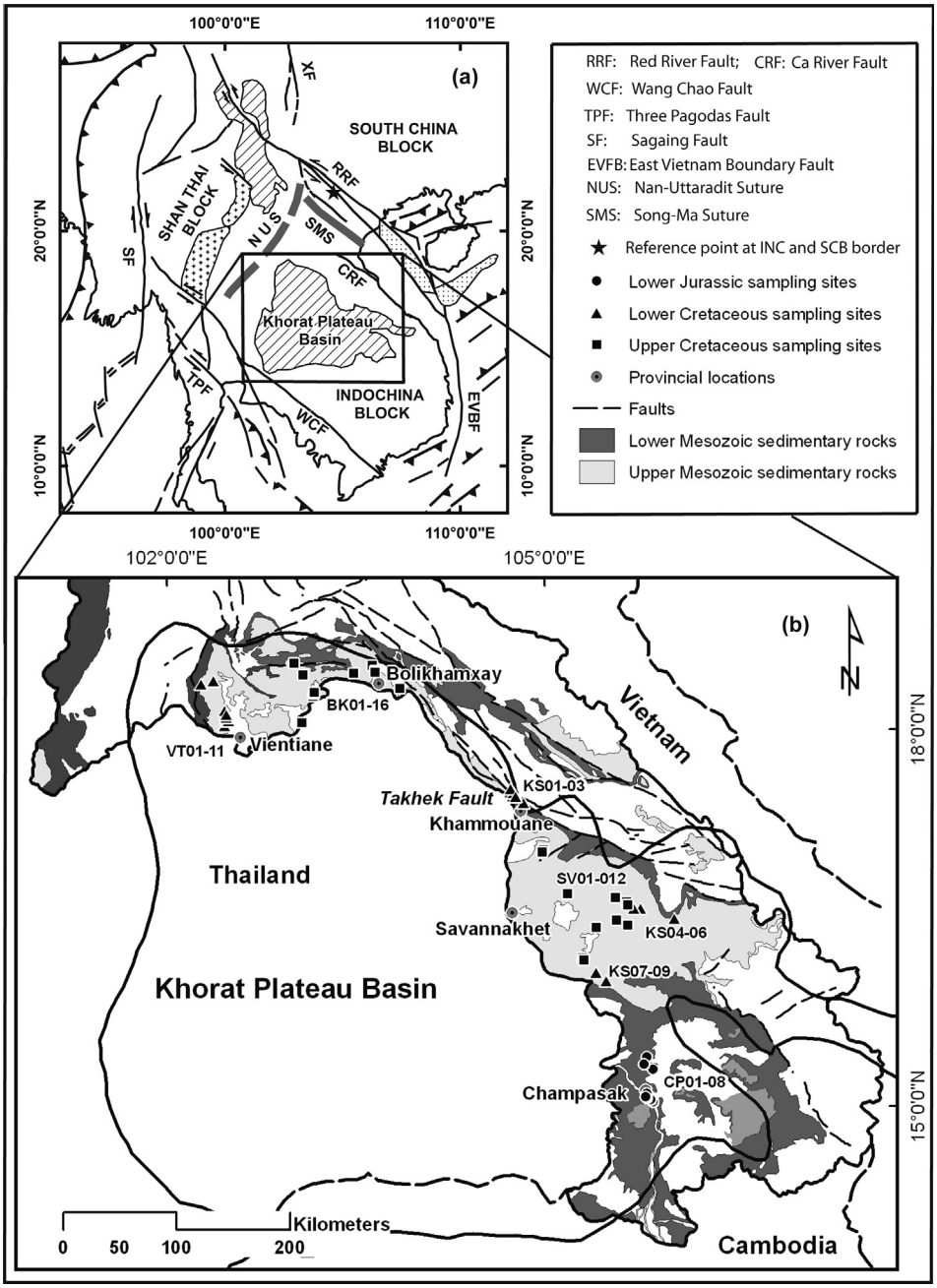


Fig. 1. (a) Simplified tectonic map of the Southeast Asia showing main blocks, sutures, faults, and Khorat Plateau (modified from Leloup et al., 1995; Otofuji et al., 2012). (b) Geologic map of the study area modified from Geological and mineral resource map of Laos; 1/1,000,000 (Department of Geology and Mines, 1991) including Khorat Plateau, localities, and sampling sites. Rectangular, triangles and squares denote the sampling sites of Lower Jurassic, Lower Cretaceous and Upper Cretaceous, respectively. RRF: Red River Fault, WCF: Wang Chao Fault; TPF: Three Pagodas Fault, SF: Sagaing Fault, CRF: Song Ca River Fault, EVFB: East Vietnam Boundary Fault, NUS: Nan-Uttaradit Suture, and SMS: Song Ma Suture.

directions of the K_{min} axes perpendicular to the subhorizontal bedding planes, whereas the directions of the K_{max} and K_{int} axes are girdle distributed within the bedding plane. This also suggests that the plate-like hematite originates onto the bedding plane and carries the primary remanence at the time of rock formation. However, some negative values of the shape parameter (T), i.e. prolate magnetic fabric, most probably rod-shaped magnetite with low degree of anisotropy have been observed in rocks from 4 sites (CP03, CP06, CP07 and CP08) in Lower Jurassic redbeds from Champasak and one site (KS06) in Savannakhet (Fig. 2). The K_{max} directions are well grouped for the Lower Jurassic Champasak sites, but the K_{min} and K_{int} directions are mixed and girdle distributed. This may suggest that secondary iron oxide mineral, rod-like magnetite, is formed during tectonic folding of Lower Jurassic redbeds in Champasak resulting in the direction of the K_{max} parallel to the fold axis. Remanence carried by new mineral magnetite is therefore secondary.

4.2. Magnetic carrier

Susceptibility versus temperature measurements were conducted on samples from all sites. The thermomagnetic curves of most samples are reversible (Fig. 3) during heating and cooling cycle, indicating that there has been no chemical change during magnetic treatments. For the Lower Jurassic samples there is a drop in susceptibility between 300 and 550 °C (Fig. 3a). The

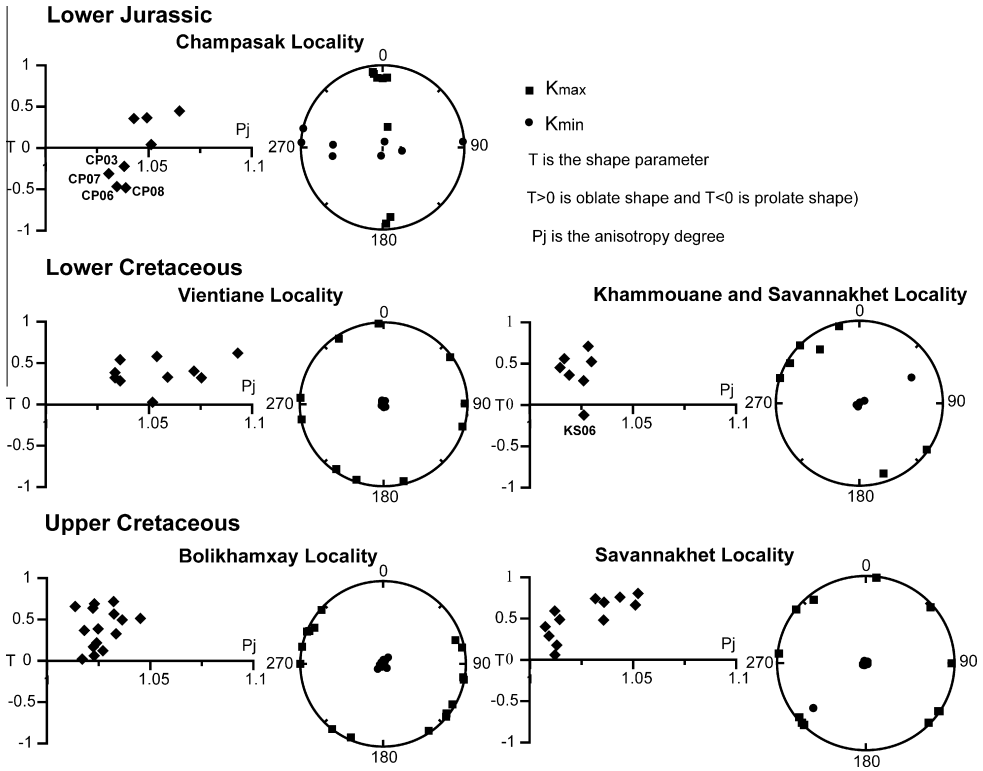


Fig. 2. *T*–*Pj* and equal area projection plots of the site mean of K_{max} and K_{min} axes on the lower hemisphere of the Mesozoic red sandstones from the central and southern Laos. *T* and *Pj*: shape parameter and corrected anisotopy proposed by Jelinek (1981). Lower Jurassic: Champasak; Lower Cretaceaous: Vientiane, and Khammouane and Savannakhet; and Upper Cretaceous: Bolikhamxay and Savannakhet.

susceptibility–temperature curve of a sample from the Lower Cretaceous redbeds shows an increase between room temperature and 500 °C, followed by a sharp decrease of susceptibility from 500 to 580 °C and a gradual decrease up to ca. 680 °C (Fig. 3b). The corresponding curve of the Upper Cretaceous redbeds (Fig. 3c) is similar to that of the Lower Jurassic, which shows a susceptibility decrease from 400 to 520 °C, followed by a gradual decrease up to about 680 °C. Such susceptibility decreasing over a temperature range may indicate variations in composition of the magnetic carriers. Based on the susceptibility changes here magnetite/titanomagnetite and hematite are probably the most dominant magnetic minerals.

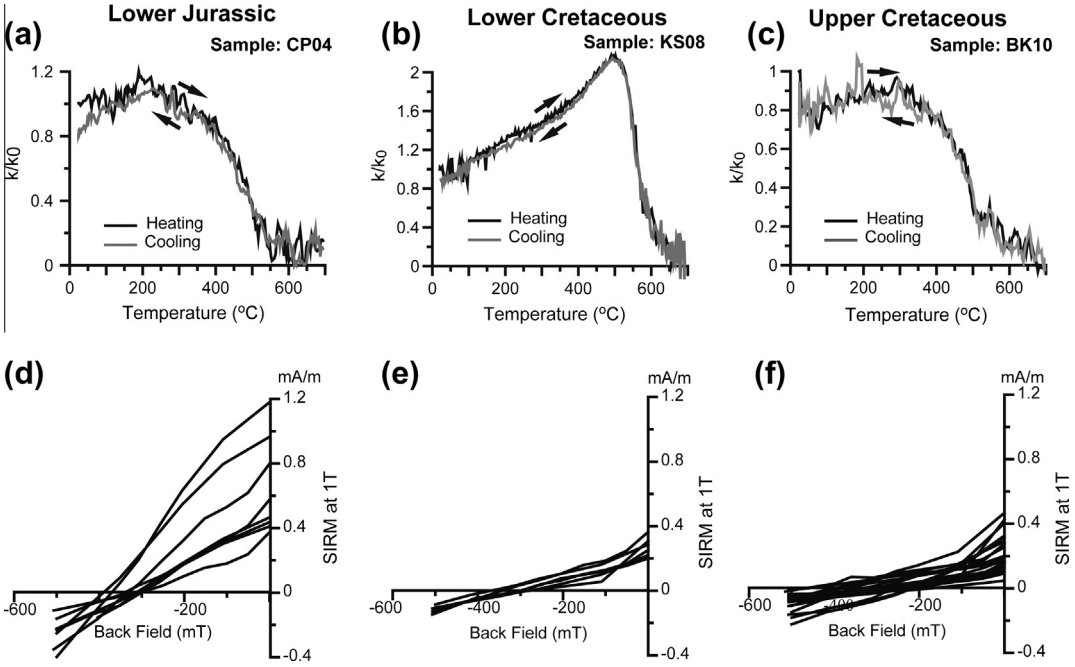


Fig. 3. (a–c): Thermomagnetic curves (K vs. T) of representative samples (CP04, SK08, and BK07) and (d–f): back-field demagnetization of IRM at 1T of representative samples for Lower Jurassic, Lower Cretaceous, and Upper Cretaceous.

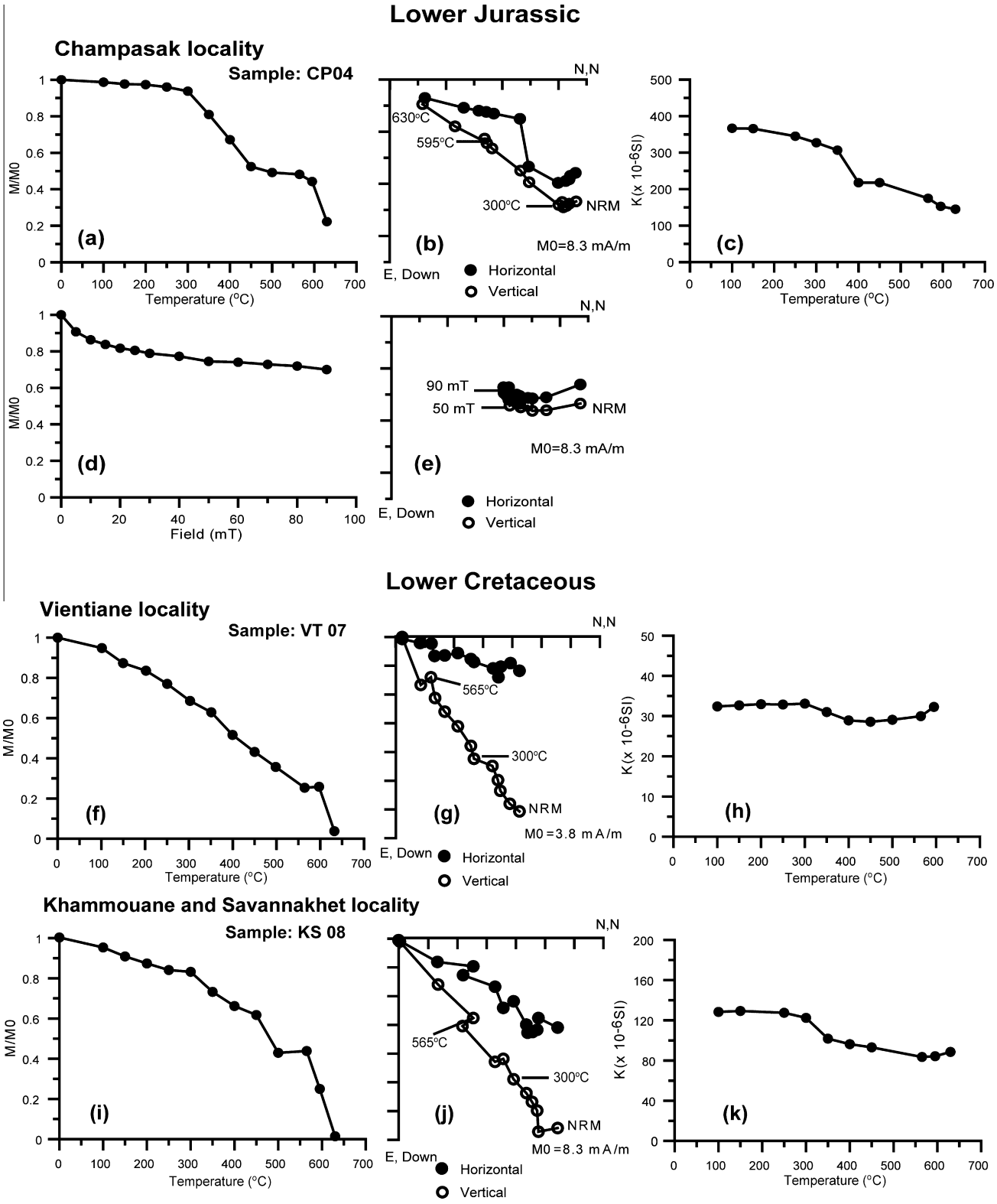


Fig. 4. Examples of remanence behaviour during progressive thermal and alternating field demagnetizations, Zijderveld plots (zijderveld, 1967), and variation of the magnetic susceptibility during thermal demagnetization of representative red sandstone samples from the study area for Lower Jurassic, Lower Cretaceous, and Upper Cretaceous times.

In the Fig. 3d–f the IRM at 1 T for Lower Jurassic, Lower Cretaceous, and Upper Cretaceous samples are shown to range between 0.4–1.2 A/m, 0.3–0.4 A/m, and 0.12–0.42 A/m, respectively. The IRM intensity for Lower Jurassic samples is thus relatively high compared to those of the Lower and Upper Cretaceous samples. Coercivity of remanence (B_{cr}), determined from the back-field demagnetization curves of IRM, ranges from 300 to 400 mT for Lower Jurassic and Lower Cretaceous and from 200 to 400 mT for Upper Cretaceous samples. High coercivity (>200 mT) samples may indicate hematite. This interpretation is supported by AF demagnetization curves (Fig. 4) which show that ca. 70% of the remanence still remain after 90 and 100 mT AF. Thermal

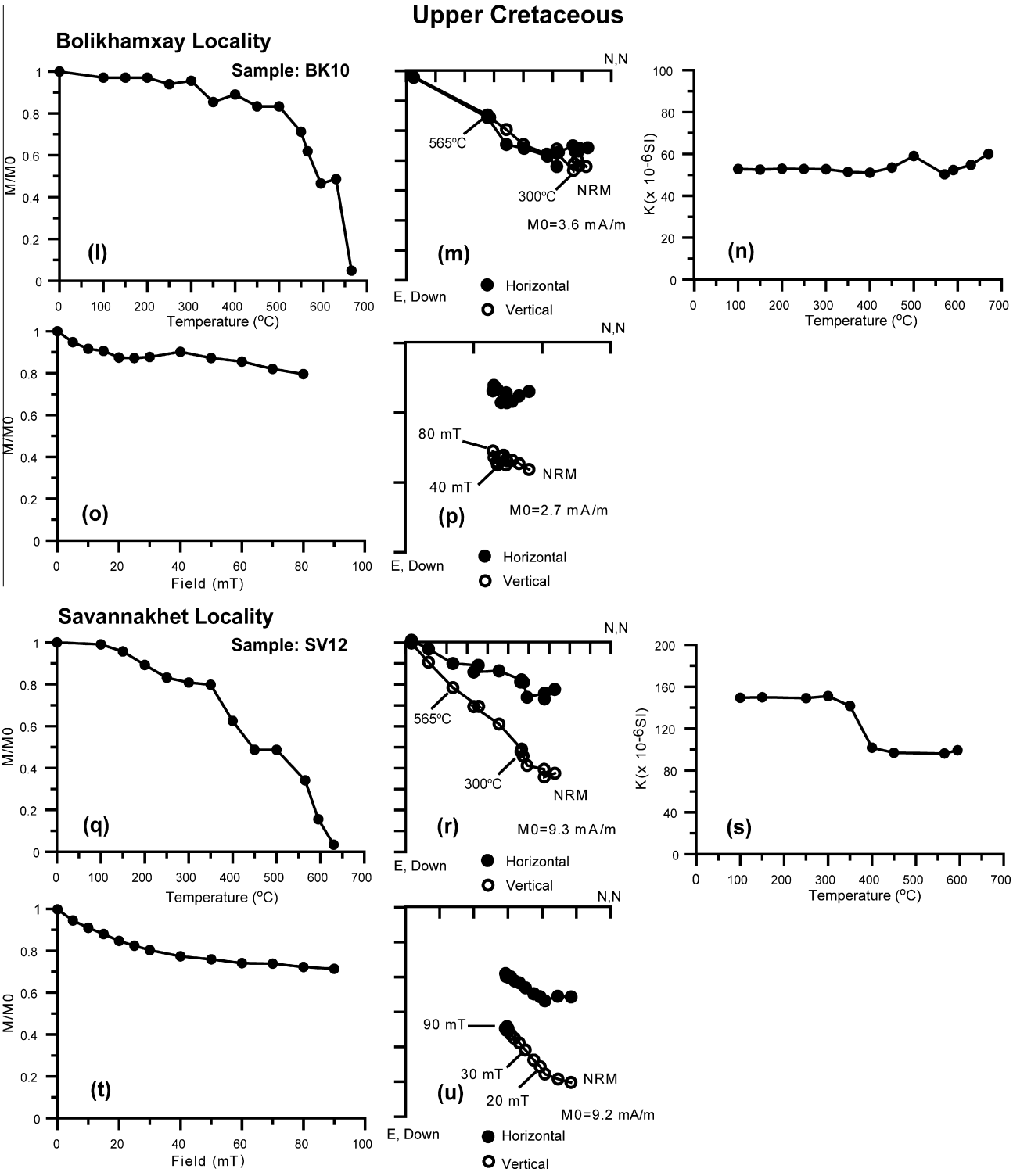


Fig. 4 (*continued*)

demagnetizations of the NRM (Fig. 4) reveal decreases of remanence in two temperature intervals, i.e. ca. 300–450 °C and 565–650 °C unblocking temperatures, probably related to titanomagnetite and/or hematite with varying unblocking temperatures. Similar results were also observed in earlier studies of rocks in the Khorat Group, NE Thailand (Yang and Besse, 1993; Bhongsuwan and Elming, 2000; Charusiri et al., 2006) and from Mesozoic sedimentary rocks from Laos showing mainly magnetite and hematite as magnetic carriers (Takemoto et al., 2009).

4.3. Palaeomagnetism

4.3.1. Lower Jurassic redbeds in Champasak

The Lower Jurassic redbeds were sampled in Champasak at Lat./Lon. = 15.2°N/105.8°E. The samples were collected in 8 sites where the dip of the bedding planes is less than 7°. Examples of thermal and AF demagnetizations are shown in Fig. 4a and d. The site mean of NRM intensity ranges from 4.2 to 10.0 mA/m. The thermal demagnetizations for most samples show unblocking temperatures

ranging between 300 and 450 °C and between ca. 600 and 630 or 670 °C. The lower temperature components (T < 300 or 350 °C) are clearly different from the medium to high temperature components and the lower declinations observed for low temperature components are probably a result from magnetic viscosity. After the removal of the low temperature component, the medium to high temperature components isolated between 350 and 595 or 630 °C are used to calculate the site mean directions. The remanence extracted from temperatures >595 or 630 °C are very small, the directions are scattered, and no characteristic remanence direction is identified. This behavior is commonly found for samples having a low NRM intensity. During AF demagnetization, a low coercivity component is eliminated after treatment in fields <20 mT. After that the remanence decreases gradually and at 90 mT AF ca. 30% of the remanent magnetization is erased (Fig. 4d). After ca. 50 mT field, a stable remanence direction could be isolated, with the vector decreasing towards the origin in the vector plot (Fig. 4e). The site mean directions are calculated using the high field components (>50 mT). The null test between the medium–high temperature component (MHTC) and high field component (HAFC) indicates that remanence carrying minerals in the rocks are probably magnetite and hematite which is no significant difference in directions; therefore, the medium–high temperature component (MHTC), temperature above 350 °C together with the high field component (HAFC) above 50 mT AF are used for calculation of site mean directions. The number of AF demagnetized and thermally demagnetized samples included in the calculation of the site mean is 55 and 37, respectively (Table 1). Formation mean directions are then calculated from the site means, which are not significantly different from each other. Only normal polarity directions are observed in this area. The formation mean directions derived from 8 sites (D_g/I_g = 34.2°/39.0°, N = 8, k_g = 244.2, α_{95} = 3.6°) before bedding corrections (Fig. 5a) is not significantly different from the mean after bedding correction (D_s/I_s = 35.2°/38.9°, k_s = 295.9, α_{95} = 3.2°; Fig. 5b). The DC fold test (Enkin, 2003) for individual sample mean directions shows the optimal clustering at 174.8 ± 37.4% unfolding, different from 0% and 100% (Fig. 5c). The inconclusive fold test is obtained at 95% confidence level for this Lower Jurassic redbeds. A Lower Jurassic mean pole is calculated at Plat./Plon. = 56.0°N/178.6°E, A_{95} = 2.6° (Tables 1 and 2; Fig. 6).

The AMS data of this Lower Jurassic redbeds are interesting. The K_{min} directions do not coincide with the bedding pole (Fig. 2) while the K_{max} well groups toward the N-S direction. The Pj–T plot of some sites, e.g. CP03, CP06, CP07, and CP08, shows predominantly prolate shape (T < 0) and the magnetic susceptibility is relatively high compare to other localities. The AMS data probably support the presence of rod-shaped magnetite (T < 0), which is newly formed during folding. It is reasonable to state that the ChRM of Lower Jurassic redbeds in Champasak is contaminated by remanence carried by the newly formed magnetite.

Inconclusive fold tests are also observed for other Khorat formations, e.g. for at Phra Wihan Formation (Bhongsuwan and Elming, 2000); Sao Khua Formation (Yang and Besse, 1993); Khok Kruat Formation (Yang and Besse, 1993; Charusiri et al., 2006), and Phu Thok Formation (Charusiri et al., 2006). Even though the fold tests are inconclusive for this study, but not 'negative', therefore, the palaeomagnetic pole is calculated and used for further analysis.

4.3.2. Lower Cretaceous redbeds in Vientiane, Khammouane, and Savannakhet

The Lower cretaceous redbeds were sampled in the Vientiane (Lat./Lon. = 18.1°N/102.4°E), and in the Khammouane and Savannakhet areas (Lat./Lon. = 16.7°N/105.4°E).

In the Vientiane area, samples were collected from 11 sites where the dip of the bedding planes are small (dip < 16°). The intensity of NRM is relatively low (0.9–4.4 mA/m). Samples from the site VT06 are rejected due to very scattered directions that may be resulted from a very low remanence intensity (NRM < 0.9 mA/m) in the range of instrumentation noise. A wide range of unblocking temperature is observed and commonly, 75% of the NRM is erased at temperatures of 565 °C and completely erased at 630 °C (Fig. 4f). Lower temperature component (T < 300 or 350 °C) shows similar behavior to those of the Lower Jurassic redbeds in Champasak; therefore, this component is rejected and not used in further analysis. The ChRMs are defined at temperatures above 350 °C after which the remanence vectors decrease towards the origin (Fig. 4g). The susceptibility monitoring shows only slightly change at the highest temperature during thermal demagnetization, which means the remanent magnetization is not affected by chemical reactions in the oven at high temperatures (Fig. 4h). The ChRMs for most samples are defined in the temperature range from 350 to 595 or to 630 °C if possible. However, for some samples there is a high unblocking temperature (T_{ub}) component isolated at temperatures higher than 595 °C or 630 °C. In two sites in this area, such high temperature components show inversed polarities of remanence. After changing the polarity, the reversed polarity site means are included in the calculation of the formation mean D_g/I_g = 21.3°/40.6°, N = 10, k_g = 10.8, α_{95} = 15.4° before bedding correction (Table 1; Fig. 5d) and D_s/I_s = 25.9/37.4°, k_s = 23.4, α_{95} = 10.2°, after bedding correction (Fig. 5e). The DC fold test (Enkin, 2003) of this locality gives optimal clustering at unfolding of 133.9 ± 50.4–100% (Fig. 5f). This positive fold test is demonstrated at 95% confidence level indicating that the primary remanence is acquired before folding. The AMS data showing the K_{min} well clustered around the bedding pole (Fig. 2) also support the presence of hematite interpreted as primary remanence carrying mineral. In earlier palaeomagnetic studies, positive fold tests were demonstrated for several rock formations in Mesozoic redbeds in SE Asia, e.g. from Khorat Group (Yang and Besse, 1993; Bhongsuwan and Elming, 2000; Charusiri et al., 2006; Takemoto et al., 2009; Otofuji et al., 2012) and from South of Vietnam (Otofuji et al., 2012). A mean palaeomagnetic pole for Lower Cretaceous redbeds from the Vientiane area is derived at Plat./Plon. = 65.1°N/171.5°E, A_{95} = 12.6° (Table 1).

In the Khammouane and Savannakhet area near Thakhek Fault zone, rocks from 9 sites with a medium to steeply dipping beds (up to 74°) were sampled. The general pattern of remanence intensity decrease during thermal demagnetization is shown in Fig. 4i. The demagnetization curves are very similar to those from the Champasak area. The remanence directions are defined from a temperature range between 350 °C and 595 °C (Fig. 4j). The susceptibility changes during heating are also small and similar to those of the samples from the Vientiane area (Fig. 4k). One sampling site (KS01) is rejected for further analysis because of a very poorly defined mean directions (α_{95} > 35°). Only normal polarity directions are found in this area. The formation mean directions of Khammouane and Savannakhet before bedding correction are derived at D_g/I_g = 30.5°/25.6°, N = 8, k_g = 11.1, α_{95} = 17.4° (Fig. 5g) and after correction at D_s/I_s = 28.8°/45.7°, k_s = 123.1, α_{95} = 5.0° (Fig. 5h). Similarly, a positive fold test (Enkin, 2003) at 95% confidence level is also obtained for this locality due to the best fitting slope at 93.3 ± 6.9–100% unfolding (Fig. 5i). This indicates that the characteristic remanence is primary and acquired before folding. A palaeomagnetic pole for this area is calculated at Plat./Plon. = 61.2°N/168.7°E, A_{95} = 5.2° (Table 1).

4.3.3. Upper Cretaceous redbeds in Bolikhamxay and Savannakhet

The Upper Cretaceous redbeds were sampled in Bolikhamxay (Lat./Lon. = 18.3°N/103.4°E) and Savannakhet (Lat./Lon. = 16.5°N/105.5°E) areas.

In Bolikhamxay area samples were collected from 16 sites where the sedimentary bedding planes dip from 2° to 37°°. The site

Table 1
Palaeomagnetic data of redbeds from central and southern Laos (this study) and from the INC and SCB from previous studies.

Site	Age	Strike/Dip	*Locality*		In situ	Tilt-corrected			Poles	
			Lat.(°N)/Lon.(°E)	n/n'	D_g(°)/I_g(°)	D_s(°)/I_s(°)	k	α_{95} (°)	Plat. (°N)/Plon. (°E)	A_{95}
Lower Jurassic (Champasak locality)										
CP01	J1-2	353/7	15.39/105.81	6/7	29.5/45.5	34.6/41.0	19.5	9.6	56.4/175.9	9.1
CP02	J1-2	267/5	15.03/105.86	5/7	33.0/38.0	31.0/33.9	43.6	6.6	60.1/184.6	5.7
CP03	J1-2	344/2	15.05/105.84	4/7	34.2/33.2	35.0/31.6	64.7	5.7	56.3/187.5	4.8
CP04	J1-2	078/6	15.06/105.81	7/7	32.5/33.9	35.6/38.1	87.3	4.3	55.7/179.8	3.9
CP05	J1-2	072/3	15.29/105.86	4/7	32.0/42.2	34.2/43.9	32.4	8.1	56.4/172.2	8.0
CP06	J1-2	085/5	15.10/105.81	7/7	36.5/35.2	39.1/38.9	53.4	5.5	52.4/179.3	5.1
CP07	J1-2	027/4	15.07/105.80	4/6	34.9/42.1	38.5/41.3	52.6	6.4	52.8/176.2	6.1
CP08	J1-2	230/5	15.33/105.79	0/7	40.7/41.3	33.6/42.3	21.4	13.4	57.2/174.2	12.9
			15.17/105.82	37/55						
		Mean (MHTC):		37	35.3/39.5					
						35.1/42.1	33.3	4.2		
		Mean (HAFC):		55	32.9/37.8		39.8	3.1		
						33.1/39.4	38.1	3.2		
			Mean (MHTC + HAFC):	*N* = 8	34.2/39.0		244.2	3.6		
				35.2/38.9	295.9	3.2	56.0/178.6		2.6	
DC Fold test for individual sample mean directions: The optimal clustering is at 95% = 174.8 ± 34.7%, different from 0 and 100% unfolding, indicating insignificant fold test										
Lower Cretaceous (Vientiane locality)										
VT01	Mz1-1	57/03	18.03/102.46	12/0	212.5/−23.6	213.7/−24.8	53.0			
(Reverse)				12/0	32.5/23.6	33.7/24.8	53.0	6.0	57.2/196.4	4.7
VT02	Mz1-1	221/14	18.02/102.47	6/0	10.1/44.8	24.2/50.9	42.9	10.3	64.3/156.0	10.1
VT03	Mz1-1	187/03	18.02/102.46	8/0	38.5/38.3	36.4/39.9	12.7	16.2	55.6/178.4	15.1
VT04	Mz1-1	063/08	18.02/102.46	9/0	355.4/39.5	358.1/43.2	8.2	19.1	50.7/89.9	16.6
VT05	Mz1-1	342/05	18.14/102.46	6/0	309.4/36.1	358.8/34.6	15.3	17.7	50.0/88.6	15.7
VT06[a]	Mz1-1	304/08	18.06/102.48	5/0	200.0/37.8	198.2/45.5	26.2	15.2	−41.6/80.6	15.4
VT07	Mz1-1	353/07	18.06/102.49	12/0	19.2/38.6	23.9/35.3	40.4	6.9	67.3/185.0	6.1
VT08	Mz1-1	247/08	18.09/102.47	4/0	232.6/−17.0	230.4/−14.8	40.4			
(Reverse)				4/0	52.6/17.0	50.4/14.8	22.9	19.5	40.3/198.0	14.4
VT09	Mz1-2	112/16	18.38/102.37	6/0	41.6/37.1	39.0/43.7	31.5	12.1	53.2/173.8	11.9
VT10	Mz1-1	266/12	18.38/102.37	9/0	26.9/49.4	21.4/38.8	27.8	9.9	69.9/178.8	9.1
VT11	Mz1-1	310/11	18.35/102.27	10/0	23.2/47.7	25.9/37.1	110.9	4.6	65.5/182.4	4.2
Mean before inversion			18.12/102.44	87/0	0.3/50.5		2.2	43.1		
Mean after inversion				*N* = 10	21.3/40.6		10.8	15.4		
			Mean (MHTC) *N* = 10		25.9/37.4	23.4	10.2		65.1/171.5	12.6
DC Fold test for individual sample mean directions: The optimal clustering is at 95% = 133.9 ± 50.4–100% unfolding, indicating positive fold test										
Lower Cretaceous (Khammouane + Savannakhet locality)										
KS01[a]	Mz2-1	139/76	17.49/104.75	8/0	150.5/9.8	148.7/−12.1	3.4	35.6	57.4/358.1	25.8
KS02	Mz2-1	129/25	17.52/104.73	8/0	25.1/20.8	20.5/44.9	44.4	8.4	69.0/165.8	8.4
KS03	Mz2-1	139/43	17.41/104.83	12/0	33.2/6.3	25.0/49.2	56.7	5.4	63.9/161.2	6.3
KS04	Mz2-1	140/74	17.46/104.77	11/0	43.2/−27.2	41.0/47.3	13.2	13.1	50.9/170.9	13.7
KS05	Mz2-1	138/13	16.57/105.76	15/0	30.5/38.3	26.2/50.6	71.7	4.5	62.0/159.2	5.0
KS06	Mz2-1	204/06	16.49/106.03	14/0	29.8/37.5	25.1/37.9	180.4	2.9	65.8/180.7	2.6
KS07	Mz1-2	010/07	16.06/105.41	11/0	25.4/41.9	31.2/39.7	67.7	5.6	59.9/178.0	5.2
KS08	Mz1-2	095/14	15.99/105.49	13/0	30.3/40.8	37.6/53.1	124.5	3.7	51.9/160.8	4.3
KS09	Mz2-1	085/03	16.57/105.71	6/0	24.6/38.5	25.9/41.1	17.1	7.0	64.7/175.4	6.6
			16.65/105.43	98/0	30.5/25.6		11.1	17.4		
			Mean (MHTC)	*N* = 8		28.8/45.7	123.1	5.0	61.2/168.7	5.2
DC Fold test for individual sample mean directions: The optimal clustering is at 95% = 93.3 ± 6.9–100% unfolding, indicating positive fold test										

(*continued on next page*)

Table 1 (*continued*)

Site	Age	Strike/Dip	*Locality*		In situ	Tilt-corrected			Poles	
			Lat.(°N)/Lon.(°E)	n/n'	D_g(°)/I_g(°)	D_s(°)/I_s(°)	k	α_{95} (°)	Plat. (°N)/Plon. (°E)	A_{95}
Upper Cretaceous (Bolikhamxay locality)										
BK01	Mz2-2	325/6	18.26/103.16	15/0	30.7/40.1	32.5/34.6	40.4	6.3	59.2/186.4	8.3
BK02	Mz2-2	114/14	18.52/103.01	10/0	33.4/34.1	35.6/47.9	24.0	10.1	55.9/168.5	15.8
BK03	Mz2-2	205/4	18.43/103.08	12/0	24.4/33.3	21.6/32.0	139.2	3.7	69.4/193.4	5.1
BK04	Mz2-2	182/10	18.44/103.48	5/0	43.0/19.6	39.8/25.9	18.6	18.2	51.5/194.6	22.4
BK05	Mz2-2	218/12	18.28/103.17	9/3	9.5/33.7	3.4/27.4	11.4	14.2	85.0/241.7	17.6
BK06	Mz2-2	218/12	18.28/103.17	12/0	15.6/37.1	8.1/31.8	19.2	10.2	82.2/199.7	13.2
BK07	Mz2-2	210/12	18.28/103.17	2/6	4.6/26.7	359.9/21.1	25.2	14.5	82.6/283.9	16.7
BK08	Mz2-2	275/02	18.29/103.17	11/0	15.9/39.3	15.7/37.4	14.9	11.1	75.0/180.5	15.2
BK09	Mz2-2	110/10	18.44/103.48	12/0	47.1/37.5	51.5/46.3	36.2	7.3	42.1/172.5	11.2
BK10	Mz2-2	037/13	18.47/103.65	11/0	25.9/32.7	34.5/34.3	66.2	5.7	57.4/187.4	7.5
BK11	Mz2-2	343/6	18.05/103.65	11/0	29.4/42.0	32.7/37.6	17.7	11.1	59.1/182.6	15.2
BK12	Mz2-2	276/12	18.05/103.63	11/0	34.0/38.5	30.5/28.1	28.7	8.7	60.6/195.1	10.8
BK13	Mz2-2	334/20	18.46/103.65	10/0	27.7/48.8	36.7/31.1	21.1	10.8	55.1/189.9	13.9
BK14	Mz2-2	183/15	18.05/103.07	10/0	18.1/41.1	4.5/43.3	6.4	20.7	81.7/132.5	30.3
BK15	Mz2-2	188/14	18.45/103.65	9/0	10.5/52.1	353.0/50.5	16.5	13.0	75.7/78.6	21.1
BK16	Mz2-2	182/37	18.32/103.85	8/5	30.3/37.4	356.7/45.4	56.7	5.6	80.9/84.9	8.4
			18.32/103.38	157/14	25.2/37.7		43.5	5.7		
			Mean (MHTC):	157	25.9/38.8		16.1	2.9		
							26.2/35.5	12.3	3.4	
			Mean (HAFC):	14	10.1/29.5		19.8	9.5		
						10.46/10.4	24.7	8.5		
			Mean (MHTC + HAFC):	*N* = 16		21.2/37.1	24.2	7.7	69.9/181.5	8.2
DC Fold test for individual sample mean directions: The optimal clustering is at 95% = 89.7 ± 19.1–100% unfolding, indicating positive fold test.										
Upper Cretaceous (Savannakhet locality)										
SV01	Mz2-2	355/06	17.04/104.99	10/0	29.1/42.5	33.3/38.9	27.18	9.4	58.3/180.6	13.1
SV02	Mz2-2	350/06	17.02/104.98	9/0	10.1/42.6	24.5/39.3	51.72	7.2	66.4/178.2	10.0
SV03	Mz2-2	340/02	16.69/105.18	5/5	29.0/36.2	29.9/34.7	26.46	9.6	61.5/185.8	12.7
SV04	Mz2-2	017/07	16.66/105.56	4/8	24.7/33.5	29.1/32.3	217.8	2.9	62.2/189.4	3.8
SV05	Mz2-2	000/06	16.62/105.65	7/7	25.7/38.1	29.7/35.3	40.5	6.3	61.7/185.3	8.4
SV06	Mz2-2	307/11	16.60/105.66	10/5	27.5/39.5	28.6/28.6	82.1	4.1	62.5/194.4	5.1
SV07	Mz2-2	204/10	16.42/105.41	6/7	18.6/32.6	12.4/31.1	37.3	8.3	78.1/191.9	10.7
SV08	Mz2-2	055/18	16.42/105.41	0/7	11.1/28.8	20.8/40.2	63.1	7.7	69.4/173.8	10.9
SV09	Mz2-2	153/08	16.48/105.57	6/7	31.0/36.6	27.3/43.2	87.1	4.5	63.1/172.2	6.6
SV10	Mz2-2	077/05	16.4/105.66	8/6	25.0/41.3	28.0/45.2	87.9	4.3	62.1/169.3	6.5
SV11	Mz2-2	013/05	16.18/105.30	8/7	30.6/48.0	35.7/46.3	63.3	4.8	55.1/170.0	7.3
SV12	Mz2-2	010/07	16.16/105.31	8/7	28.8/43.7	34.7/41.1	242.7	2.5	56.6/176.8	3.5
			16.47/105.47	81/66						
			Mean (MHTC):	81	27.6/39.4		38.1	2.6		
						29.5/42.3	30.0	3.0		
			Mean (HAFC):	66	24.1/38.5		47.4	2.6		
						24.0/41.7	32.0	3.2		
			Mean (MHTC + HAFC):	*N* = 12	24.9/38.8		128.0	3.9		
						27.7/38.0	114.1	4.1	63.3/180.0	3.9
DC Fold test for individual sample mean directions: The optimal clustering is at 95% = 93.1 ± 20.4–100% unfolding, indicating positive fold test.										

Note: Lat.(°N) and Lon.(°E): latitude and longitude; respectively; N: number of sites measured and used for calculation; n: Thermal demagnetized samples; n': AF demagnetized samples; D_g and I_g (D_s and I_s): declination and inclination in situ (declination and inclination after tilt corrected), respectively; k_g and k_s: the estimate of the precision parameter (Fisher, 1953); α_{95} is the radii of the cone of 95% confidence; A_{95}, the radius; that the mean pole lies within 95% confidence.

[a] Rejected sites for calculation of the mean formation; J1-2: Lower Jurassic; Mz1-2 and Mz2-1: Lower Cretaceous; Mz2-2: Upper Cretaceous; MHTC: Medium – High Temperature Component; and HAFC: High AF Component.

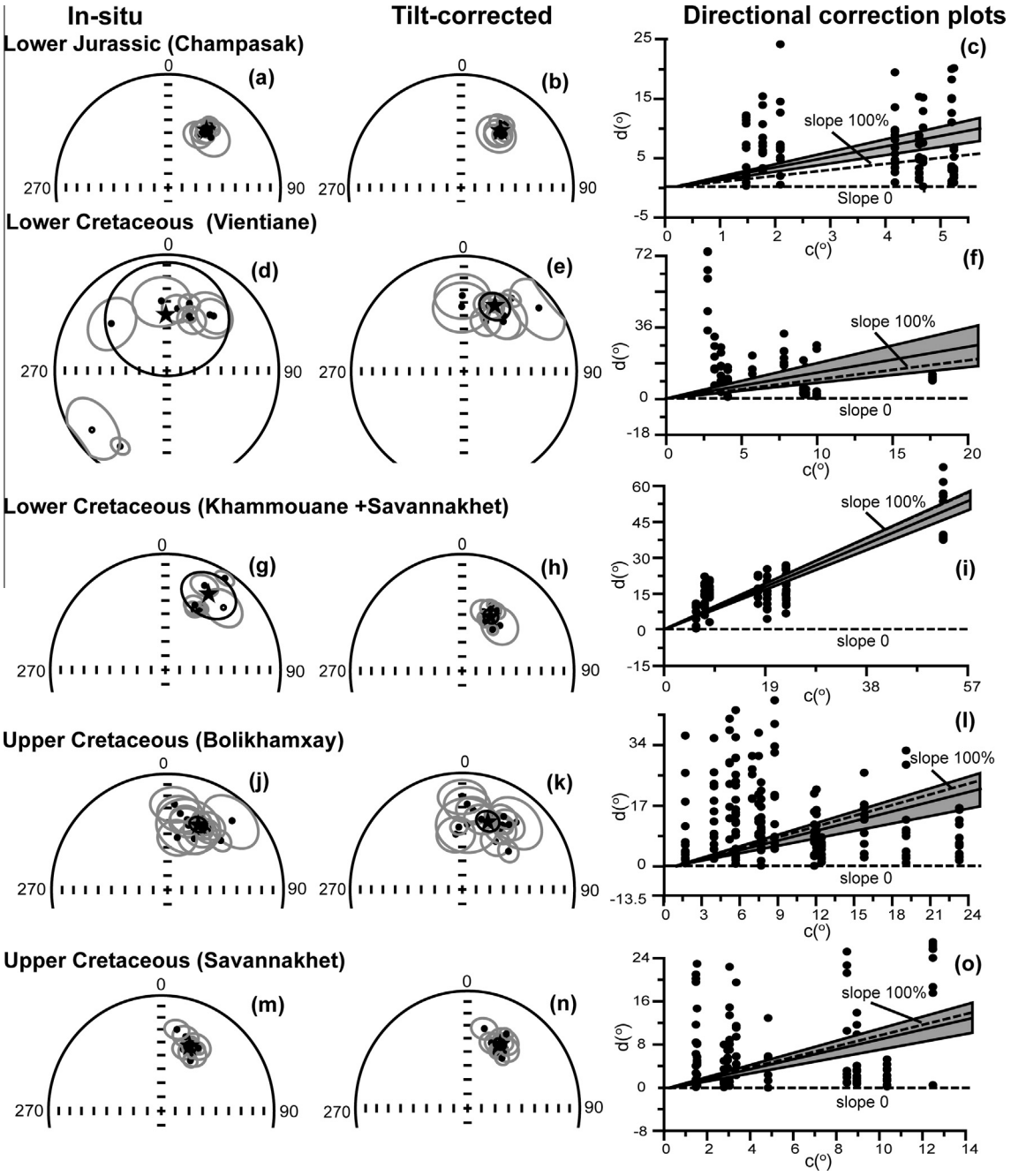


Fig. 5. Lower hemisphere equal area stereographic projection plots of site mean remanence directions for high temperature and alternating field together with the 95% cone of confidence before and after tilted correction for Lower Jurassic: Champasak; Lower Cretaceous: Vientiane, and Khammouane and Savannakhet; Upper Cretaceous: Bolikhamxay and Savannakhet. Solid and opened squares are the normal and reversed polarities, respectively. The direction-correction (DC) fold test of Enkin (2003) plots show optimal clustering of precision parameter (shading area) together with DC slopes 0% and 100% (dash lines), and individual sample means (black dots) for each locality. The stars denote the mean directions of the localities.

mean of NRM intensities ranges between 1.0 and 9.2 mA/m. The T_{ub} ranges from 350 °C to 670 °C and ca. 50% of the remanence is erased at 565 °C, and almost 100% erased at 670 °C (Fig. 4l). After the removal of low temperature magnetizations by 350 °C, the remanence decreases consecutively towards the origin from 350 to 595 °C or 630 °C if possible (Fig. 4m). These components isolated in this temperature range are used to calculate the site mean directions. The heating procedure does not affect the remanent magnetization since the change in susceptibility is small (Fig. 4n). During AF demagnetization a low to medium coercivity remanence was erased in AF < 40 mT. After 90 mT AF, ca. 80% of the remanence still remain (Fig. 4o). High-field components (>40 mT), with vectors decreasing towards the origin, are used to calculate the site mean directions. For this locality, the number of AF demagnetized samples and thermally demagnetized samples used for the mean calculations is 14 and 157, respectively. The formation mean directions, based on vectors defined from thermal and AF demagnetizations samples (MHTC + HAFC) are well defined at D_g/I_g = 25.2°/37.7° (N = 16, k_g = 43.5, α_{95} = 5.7°; Fig. 5j) and not significantly different from the mean defined after bedding correction (D_s/I_s = 21.2°/37.1°, k_s = 24.2, α_{95} = 7.7°; Fig. 5k). The DC fold test (Enkin, 2003) is also positive at 95% confidence level for these localities (the best fitting slope at 89.7 ± 19.1–100% unfolding, Fig. 5l), indicating that occurrence of primary remanence is taken

Table 2
Summarized palaeomagnetic data sets from the INC and SCB.

Formation/locality	Lat. (°N)/Lon. (°E)	Age	*N*/*n*	Fold test	D_s/I_s	*k*	α_{95} (°)	Plat. (°N)/Plon. (°E)	A_{95} (dp/dm)	References
Indochina block										
Lower Triassic–Middle Jurassic										
Phu Kradung Formation		Jl-Jm	5/–	–	30.6/31.0	60.8	9.9	60.7/182.8	10.6	Bhongsuwan and Elming (2000)
Phra Wihan Formation		Jm	11/–	–	30.3/27.6	97.5	4.6	60.7/187.4	4.0	Bhongsuwan and Elming (2000)
Sao Khua Formation		Ju	5/–	–	30.4/45.1	59.1	10.0	59.9/173.7	10.5	Bhongsuwan and Elming (2000)
Nam Phong Formation		Tru–Jl	–/61	–	37.2/40.1	71.6	6.6	54.4/175.6	–	Yang (1992)
Mean			4/–		32.7/36.6	115.6	7.1	59.1/179.6	5.1	
Champasak	15.17/105.82	Jl	8/92	Insignificant	35.2/38.9	259.8	3.2	56.0/178.6	2.6	This study
Null test for directions	(poles): $F = 2.2 < F(2, 20) = 5.72$ ($F = 0.0 < F(2, 20) = 5.72$). It is not significant difference between this and previous directions (poles) at 95% confidence level.									
Upper Jurassic–Mid Cretaceous										
Bolikhamxay	18.5/103.8	J3–Kl	8/–	Positive	42.1/42.9		7.9	50.7/169.7	8.7	Takemoto et al. (2009)
Maung Phin	16.5/106.0	J(Mid K)	23/–	Syn-folding	30.8/39.9	–	3.0	60.5/178.6	3.0	Takemoto et al. (2009)
Phu Phan-Khok Kruat	–	Kl	3/–	Insignificant	30.2/36.1	108.0	11.9	61.1/182.1	8.1	Bhongsuwan and Elming (2000)
Nam Nao	16.5/103.0	Kl	10/–	Insignificant	28.1/40.5		2.4	62.7/173.3	2.4	Yang and Besse (1993)
Maung Sakon Nakhon	16.5–17.2/102.5–104.1	Kl	–/4	Positive	31.8/38.3		5.7	59.7/182.7	5.7	Charusiri et al. (2006)
Phu Phan-Khok Kruat	–	Kl–km	–/42	–	31.4/27.1	27.0	9.4	59.7/192.7	–	Charusiri et al. (2006)
Phu Phan-Khok Kruat	–	Kl–km	–/77	–	32.0/40.5	390.6	2.4	62.7/173.3	–	Yang (1992)
Phu Phan	–	Kl	–/9	–	45.0/19.0	10.0	41.0	46.0/179.0	–	Maranate and Vella (1986)
Sao Khua	–	Kl	–/48	–	33.0/35.0	23.0	8.0	59.0/184.0	–	Maranate and Vella (1986)
Phra Wihan	–	Ju–Kl	–/42	–	31.0/41.1	10.0	13.0	60.0/174.0	–	Maranate and Vella (1986)
Phu Kradung-Sao Khua	–	Ju–Kl	–/73	–	26.6/37.3	443.3	2.3	64.0/187.1	–	Yang (1992)
Mean			11/–		33.0/36.7	84.3	5.0	58.9/179.5	3.7	
Vientiane	18.12/102.44	Kl	10/87	Positive	25.9/37.4	23.38	10.2	65.1/171.5	12.6	This study
Khammouane + Savannakhet	16.65/105.43	Kl	8/98	Positive	28.2/45.5	11.9	5.3	61.2/168.7	5.2	This study
Vientiane and Khammouane + Savannakhet	17.19/104.29		18/185		27.2/41.2	34.7	4.4	63.3/170.2	6.9	This study
Null test for directions	(poles): $F = 1.77 < F(2, 54) = 3.17$ ($F = 1.93 < F(2, 54) = 3.17$). It is not significant difference between this and previous directions (poles) at 95% confidence level.									
Middle-Upper Cretaceous										
*Lai Chau	22.3/103.4	K3	5/–	Positive	12.2/40.1		4.7	78.7/188.0	5.1	Takemoto et al. (2005)
*Yen Chau	21.0/104.4	K3	8/–	Positive	3.2/26.7	12.9	12.9	83.2/255.6	10.8	Takemoto et al. (2005)
Phu Thok	–	km–Ku	–/189	–	31.8/28.7	111.4	3.5	59.4/190.8		Charusiri et al. (2006)
Khok Kruat	–	km	–/21	–	47.0/29.0	19.0	14.0	45.0/189.0		Maranate and Vella (1986)
Maung Sakon Nakhon	16.5–17.2/102.5–104.1	K2	8/–	–	31.4/27.1		9.4	59.7/192.7	9.1	Takemoto et al. (2005)
*Da Lat	11.7/108.3	Ku	2/–	Positive	11.4/35.4	331.7	1.7	76.5/161.2	1.7	Otofuji et al. (2012)
Mean			3/–		36.7/28.5	105.8	5.0	58.9/179.5	12.0	
Bolikhamxay	18.32/103.38	Ku	16/171	Positive	19.7/37.4	23.5	7.5	69.9/181.5	8.2	This study
Savannakhet	16.47/105.47	Ku	12/147	Positive	25.6/38.0	103.5	4.3	63.3/180.0	3.9	This study
Bolikhamxay and Savannakhet	17.34/	104.38	28/318		24.0/27.3	36.2	4.6	67.0/180.8	4.9	This study
Null test for directions	(poles): $F = 2.6 < F(2, 58) = 3.16$ ($F = 2.36 < F(2, 58) = 3.16$). It is not significant difference between this and previous directions (poles) at 95% confidence level									
South China Block										
Lower Jurassic										
		Jl	–	–	–	–	–	69.4/192.4	–	Enkin et al. (1992)
Sichuan	26.8/102.3	Jl	–	–	–	–	–	63.8/188.2	16.1	Huang and Opdyke (1992)
Guangxi	22.0/108.7	Jl	–	–	–	–	–	78.0/251.8	6.3	Gilder et al. (1993)
Guangxi	22.6/108.5	Jl	–	–	–	–	–	79.0/225.8	14.0	Gilder et al. (1993)
Guangxi	22.6/108.5	Jl	–	–	–	–	–	60.4/204.5	14.0	Gilder et al. (1993)
					Mean pole (5 poles)			71.3/205.4	10.7	
Lower Cretaceous										
–	27.9/102.3	Kl	2/11	–	–	–	–	77.4/196.2	11.5/18.3	Zhu et al. (1988)
–	27.9/102.3	Kl	7/–	–	–	–	–	69.0/204.6	4.3	Huang and Opdyke (1992)
–	30.0/102.9	Kl	23/–	–	–	–	–	74.5/229.0	2.7/4.7	Enkin et al. (1991)
–	25.7/101.2	Kl	–	–	–	–	–	78.8/194.8	3.7	Liang (1990) In: Cheng et al. 1996)
–	30.1/103.0	Kl	8/–	–	–	–	–	78.6/201.0	15.3	Zhuang et al. (1988)
–	19.6/109.5	Kl	–	–	–	–	–	80.0/246.4	–	Yu et al. (1990)

Table 2 (*continued*)

Formation/locality	Lat. (°N)/Lon. (°E)	Age	*N*/*n*	Fold test	D_s/I_s	*k*	α_{95} (°)	Plat. (°N)/Plon. (°E)	A_{95} (dp/dm)	References
–	–	Kl	7/–	–	–	–	–	77.1/227.6	5.5	Lin (1984)
–	–	Kl	–/305	–	3.2/26.7	33.3	–	71.0/219.8	4.1	Enkin (1990)
Zhejiang Province		Kl	19/–	–	9.6/47.6	37.6	5.5	81.8/207.1	6.9	Morinaga and Liu. (2004)
Nanning	22.7/107.8	K1	8/–	Positive	352.8/38.2		5.5	86.5/26.4	6.8	Gilder et al. (1993)
Jiangxi	26.9/115.3	K1	23/–	Positive	12.9/38.6		3.4	76.3/224.3	3.3	Tsuneki et al. (2009)
Nanning (East)	22.4/108.4	K1	12/–	Positive	10.8/39.1		9.0	80.8/193.3	9.2	Kawamura et al. (2013)
Nanning	22.6/108.5	K1	29/–	Positive	3.3/40.2		4.4	87.2/177.6	4.6	Kawamura et al. (2013)
Nanning (West)	22.4/108.4	K1	17/–	Positive	351.8/40.7		3.8	87.9/50.3	4.0	Kawamura et al. (2013)
Hainan	19.0/109.2	K1	6/–	positive	4.3/41.8		9.4	83.2/143.0	9.8	Li et al. (1995)
Hainan	19.0/109.2	K1	17/–	54% untilted	4.3/48.4		5.4	77.7/162.1	4.4	Liu and Morinaga, 1999
					Mean Pole (for 16 Poles):			80.7/206.6	3.6	
Middle-Upper Cretaceous										
–	26.6/102.4	K2	–	–	–	–	–	78.9/186.6	4.3/6.7	Zhu et al. (1988)
–	26.5/102.4	K2	18/–	–	–	–	–	81.5/220.9	7.1	Huang and Opdyke (1993)
–	32.0/119.0	K2	10/–	–	–	–	–	76.3/172.6	10.3	Kent et al. (1986)
–	27.0/115.0	K2	–	–	–	–	–	79.2/158.6	1.0	Gu and Gu (1990) In: Ma et al. (1993)
–	23.0/115.0	K2	6/–	–	–	–	–	66.0/221.5	3.0	Hsu (1987)
–	19.6/109.5	K2	–	–	–	–	–	80.6/175.1	–	Yu et al. (1990) In: Ma et al. (1993)
–	27.9/102.3	K2	8/–	–	–	–	–	80.5/187.3	–	Zhu In: Ma et al. (1993)
–	25.7/101.2	K2	–	–	–	–	–	74.2/188.0	–	Liang (1990) In: Cheng et al. (1996)
–	29.9/103.1	K2	7/–	–	–	–	–	83.0/188.7	8.4	Zhuang et al. (1988)
–	29.9/103.1	K2	8/–	–	–	–	–	80.4/208.2	7.3	Zhuang et al. (1988)
–	30.8/118.2	K2	10/-	–	–	–	–	83.8/200.3	14.6	Gilder et al. (1999)
–	30.0/102.9	K2	–	–	–	–	–	72.8/214.1	6.6	Enkin et al. (1991)
Fexiangua area	–	Ku	–	–	15.0/30.8	38.0	6.1	70.8/234.0	–	Enkin (1990)
Nanjing	–	km–Ku	–	–	–	42.0	8.0	76.0/173.0	–	Kent et al. (1987)
Zhejiang Province	28.4–29.1/119.6–120.0	K2	19/–	Positive	10.4/45.5	36.2	5.7	81.0/214.3	6.4	Morinaga and Liu (2004)
Fujian Province	25.1–26.4/116.4–117.8	K2	22/–	Positive	11.5/44.2	35.3	5.3	79.7/201.7	5.8	Morinaga and Liu (2004)
Guangdong Province	24.1–24.2/115.2–115.8	K2	9/–	Positive	10.7/44.0	42.8	8.0	79.8/188.3	7.7	Morinaga and Liu (2004)
Jishui	27.1/115.1	K2	10/–		355.7/34.8		6.3	81.0/322.2	5.8	Wang and Yang (2007)
					Mean Pole (for 18 Poles):			79.6/200.9	3.2	

Note: Lat.(°N)/Lon.(°E): latitude/longitude of sites; N/n: number of sites/samples; D_s/I_s: declination/inclination in stratigraphic coordinate; α_{95}: the radii of the cone of 95% confidence; Plat.(°N)/Plon.(°E): pole latitude/pole longitude; A_{95}, the radius; that the mean pole lies within 95% confidence; dp and dm: ellipse of A_{95} per cent confidence along the site-to-pole great circle and perpendicular to the great circle, respectively; Tru-Jl: Upper Triassic–Lower Jurassic; Jl: Lower Jurassic; Jm: Middle Jurassic; Jml: Middle-Lower Jurassic, J(Mid K): Jurassic-Middle Cretaceous; J3-K: Upper Jurassic-Cretaceous; Ju-Kl: Upper Jurassic-Lower Cretaceous; Kl: Lower Cretaceous; Kl-km: Lower-Middle Cretaceous; Kl: Lower Cretaceous; K2: Middle-Upper Cretaceous. (*): the data are excluded for calculation of the mean directions and mean poles.

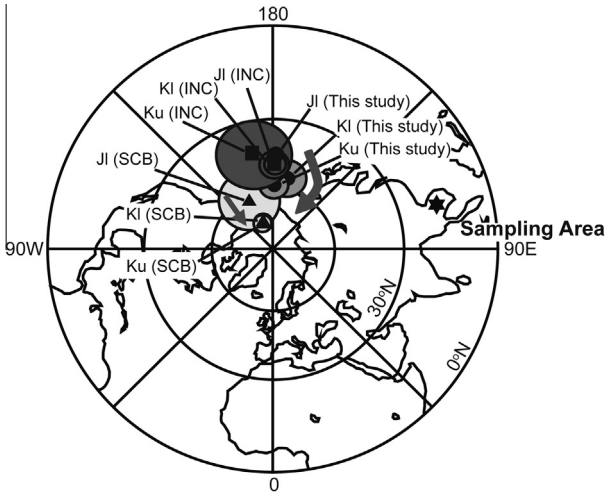


Fig. 6. Palaeomagnetic pole positions together with 95% confidence circle for the INC and SCB during Lower Jurassic to Upper Cretaceous. Jl, Kl, and Ku denote Lower Jurassic, Lower Cretaceous, and Upper Cretaceous, respectively. INC: Indochina Block and SCB: South China Block. Gray arrows denote trend of APWP for INC and SCB from Lower Jurassic to Upper Cretaceous time. The star is the location of study area.

place before folding. A mean palaeomagnetic pole for Upper Cretaceous is calculated from the bedding corrected formation mean at Plat./Plon. = 69.9°N/181.5°E, A_{95} = 8.2° (Table 1).

The 12 sites sampled in Savannakhet locality reveal site means of NRM intensities between 2.2 and 13.2 mA/m. Thermal demagnetization curves show (Fig. 4q) unblocking temperatures from 350 to 450 °C and 500 to 630 °C. At temperature of 565 °C, about 20% of the remanence is left and at 630 °C it is totally erased. After 90 mT, AF demagnetization ca. 30% of the remanence is erased (Fig. 4t). The site mean directions are calculated from high coercivity components (>40 mT) and unblocking temperatures from 350 to 595 or 630 °C. The proportion of AF versus thermally demagnetized samples used in the mean calculations is 66/81. A well-defined formation mean is calculated from the 12 site means at D_g/I_g = 24.9°/38.8°, (k_g = 128.0, N = 12, α_{95} = 3.9°, Fig. 5m) and the mean is not significantly different from the bedding corrected mean (D_s/I_s = 27.7°/38.0°, k_s = 114.1, α_{95} = 4.1°; Table 1; Fig. 5n). The optimal clustering at 93.1 ± 20.4–100% unfolding indicates the positive fold test at 95% confidence level (Fig. 5o). The remanence was acquired before folding of rock and could be primary. A palaeomagnetic pole for the Lower Cretaceous redbeds calculated from the formation mean after bedding correction from Savannakhet is derived at Plat./Plon. = 63.3°N/180.0°E, A_{95} = 3.9° (Table 1).

4.4. Palaeoposition and reconstructions of Indochina Block

In the last two decades, a number of palaeomagnetic studies have been focused on Indochina, Shan Thai, and South China blocks. The extrusion model of the INC is the most accepted tectonic model for Southeast Asia (Tapponnier et al., 1982, 1986; Rowley, 1996; Replumaz and Tapponnier, 2003; Replumaz et al., 2004; Ali and Aitchison, 2007, 2008). After India–Asia collision, the idea of a lateral extrusion of SE Asia along the ASRR shear zone and the conjugate right-lateral Wang Chao and Three Pagodas faults between 32 and 17 Ma has been widely accepted (Leloup et al., 1995, 2001; Harrison et al., 1996; Wang et al., 2006; Yin, 2010). The tectonic evolutions of the Indochina have been discussed on the basis of palaeomagnetic interpretations (Yang and Besse, 1993; Chi and Dorobek, 2004; Takemoto et al., 2005; Charusiri et al., 2006; Takemoto et al., 2009; Otofuji et al., 2012; Cung and Geissman, 2013) and here we add new palaeomagnetic data for tectonic reconstructions of the INC since Jurassic time. To calculate relative poleward translations and vertical axis rotations of the Indochina Block, palaeomagnetic poles of the South China Block, the adjacent plate of Indochina Block, are selected as reference poles. The Lower Jurassic to Upper Cretaceous palaeomagnetic pole positions for the Indochina and South China are shown in Fig. 6, Tables 2 and 3. A position at boundary of the INC and SCB (Lat./Lon. = 21.5°N/105.0°E) is selected to be a geographic reference position for the reconstructions (star and open circle in Fig. 7d). This position is also used for calculating the expected palaeomagnetic directions and palaeolatitudes of the reference position in the INC and SCB using the Indochina and South China palaeomagnetic poles. The method of Butler (Butler, 1992) is used to calculate the poleward and vertical axis rotation between

Table 3
Mean palaeomagnetic poles of this study calculated for the Indochina Block (INC) and reference poles for the South China Block (SCB) from Table 2.

Age (locality)	Locality	Mean poles		Reference pole		Rotation	Poleward	Expected directions and palaeolatitudes from	
	Lat. (°N)/Lon. (°E)	Plat. (°N)/Plon. (°E)	A_{95}	Plat .(°N)/Plon. (°E)	A_{95}	$R \pm \Delta R$ (°)	$p \pm \Delta p$ (°)	INC (°)	SCB (°)
Lower Jurassic (Champasak locality)	15.17/105.82	56.0/178.6	2.6	71.3/205.4	10.7	16.4 ± 9.0	−10.8 ± 8.8	I_O = 45.3 ± 3.2 D_O = 36.9 ± 2.9 λ_O = 26.8 $\lambda+$ = 29.5 $\lambda-$ = 24.3	I_O = 31.5 ± 17.0 D_O = 19.3 ± 11.2 λ_O = 17.1 $\lambda+$ = 29.5 $\lambda-$ = 7.4
Lower Cretaceous (Vientiane and Khammouane + Savannakhet)	17.19/104.29	63.3/170.2	6.9	80.7/206.6	3.6	17.8 ± 6.8	−11.1 ± 6.2	I_O = 49.3 ± 7.8 D_O = 28.2 ± 8.0 λ_O = 30.2 $\lambda+$ = 37.8 $\lambda-$ = 23.8	I_O = 35.1 ± 5.4 D_O = 9.7 ± 3.8 λ_O = 19.4 $\lambda+$ = 23.1 $\lambda-$ = 15.9
Upper Cretaceous (Bolikhamxay and Savannakhet)	17.34/104.38	67.0/180.8	4.9	79.6/200.9	3.2	13.3 ± 5.0	−5.3 ± 4.7	I_O = 43.3 ± 6.3 D_O = 24.8 ± 5.4 λ_O = 25.2 $\lambda+$ = 30.5 $\lambda-$ = 20.6	I_O = 36.2 ± 4.7 D_O = 11.0 ± 3.4 λ_O = 20.1 $\lambda+$ = 23.4 $\lambda-$ = 17.0

Note: The reference point is selected at the INC and SCB boundary (Lat./Lon. = 21.5°N/105.0°E). R and p (ΔR and Δp) are the vertical axis rotation and poleward translation (error of R and p) of INC respect to SCB, respectively. D_O/I_O is expected declination/inclination and λ_O, $\lambda+$, and $\lambda-$ are the mean, maximum, and minimum palaeolatitudes calculated from INC poles of this study.

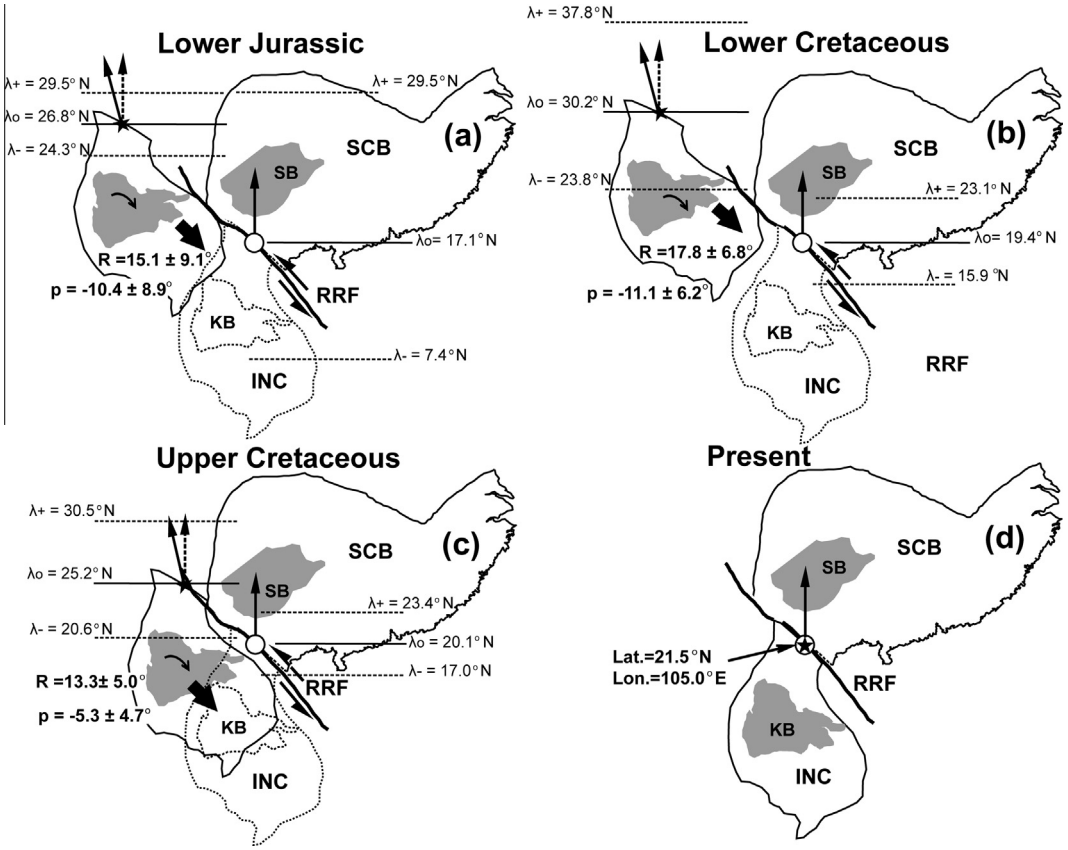


Fig. 7. Simplified reconstruction model of South China and Indochina blocks during (a) Lower Jurassic, (b) Lower Cretaceous, (c) Upper Cretaceous and (d) the Present. RRF: Red River Fault; INC: Indochina Block; SCB: South China Block; KB: Khorat Plateau Basin; and SB: Sichuan Basin. λ_O, $\lambda+$, and $\lambda-$ are the mean, maximum, and minimum palaeolatitudes calculated from the INC poles of this study whereas p and R denote the latitude translation and vertical axis rotation of INC with respect to SCB, respectively. The plate boundaries are basically modified from model of Seton et al. (2012) together with the Khorat and Sichuan basins (Leloup et al., 1995; Otofuji et al., 2012). The star and opened circles are the marked points at the INC and SCB boundary.

the INC and SCB. Our simplified tectonic model of INC and SCB from Lower Jurassic to Upper Cretaceous is created on the basis of the boundary of plates proposed by Seton et al. (2012).

4.4.1. Lower Jurassic time

The pole here calculated for the Lower Jurassic (Plat./Plon. = 56.0°N/178.6°E, A_{95} = 2.6°) is similar to other coeval poles determined from the Indochina Block (Yang, 1992; Bhongsuwan and Elming, 2000; Table 2). The null tests are passed at 95% confidence level when comparing the Champasak's pole (Table 1) with those calculated from Khorat Plateau's poles (Table 2, Yang, 1992; Bhongsuwan and Elming, 2000). It suggests that there has been no relative movement between Khorat Plateau, NE Thailand and the west of central and southern Laos or united block since Lower Jurassic time.

The expected directions, as calculated from the INC and SCB poles at the reference position, are D_o/I_o = 36.9° ± 2.9°/45.3° ± 3.2° and D_o/I_o = 19.3° ± 11.2°/31.5° ± 17.0°, respectively. These directions are significantly different from the direction of the present Earth's magnetic field (IGRF, ca. Dec./Inc. = −1.4/31.4°) and may argue for them being primary. The palaeolatitudes of the INC and SCB (with minimum and maximum latitudes calculated from A_{95}) in the Lower Jurassic as calculated from the reference poles are 26.8°N (24.35°N and 29.5°N) and 17.1°N (7.4°N and 29.5°N), respectively (Table 3). These latitudes are in the same range of palaeolatitudinal positions as for the Yunnan Basin (23.7 ± 2.5°N; Yang and Besse, 1993).

The palaeolatitude of the INC is significantly different from the present location. Therefore, the Khorat Plateau was in this time located north of its present position similar to what was suggested by Bhongsuwan and Elming (2000), Carter and Bristow (2003), Charusiri et al. (2006); and Li et al. (2004). In our reconstruction the Khorat Plateau in that time was probably located close to the Sichuan Basin in the SCB (Fig. 7a). A reconstruction is in agreement with a recently presented tectonic model proposed by Metcalfe (2013). Evidence of fossils from the Upper Permian Cathaysian *Gigantopteris* flora (Asma, 1984) was found in Yangtze platform in the South China Block and in the Indochina Block, which suggests that these blocks were joined in the Upper Permian time (Leloup et al., 1995; Metcalfe, 2002) and united together in the Middle Triassic time (Faure et al., 2014) or Upper Triassic (Metcalfe, 2013). The similarity of vertebrate fauna from the Phu Kradung Formation of Khorat Plateau and from Shaximiao Formation of Sichuan Basin was a reason to support interpretations that these areas were together (Yang and Besse, 1993; Bhongsuwan and Elming, 2000; Carter and Bristow, 2003; Li et al., 2004; Charusiri et al., 2006; Buffetaut et al., 2006).

However, it must be stated here that the palaeolongitudinal position cannot be obtained from palaeomagnetic analysis. The poleward translation and vertical axis rotation of Indochina with respect to South China are calculated as −10.8 ± 8.8° and 16.4 ± 9.0°, respectively (Table 3). By using a palaeoazimuth of ca. N45°W for the Red River Fault, a relative latitudinal displacement of the Indochina Block of about 11° can be estimated, which corresponds to a movement of 1700 km along the RRF after Lower Jurassic time. Based on palaeomagnetic data Yang and Besse (1993) suggested that major relative latitudinal translation occurred at the end of Mesozoic and then during Cenozoic times. Geological

markers found from redbed basins in Indochina and South China (Leloup et al., 1995) is realistic to advocate palaeo-location of tectonic plates involved.

4.4.2. Lower Cretaceous time

A Lower Cretaceous palaeomagnetic pole is derived from a combination of poles from Vientiane, Khammouane and Savannakhet (Table 2) at Plat./Plon. = 63.3°N/170.2°E, A_{95} = 6.9° (Tables 2 and 3; Fig. 6). There is no significant difference between the formation mean poles and palaeomagnetic poles calculated for other part of the INC (Table 2) indicating that the central part of the Indochina Block has been united as a rigid block since Lower Cretaceous. The palaeomagnetic directions at the reference position of the INC (SCB) calculated from this pole are D_o/I_o = 28.2 ± 8.0°/49.3 ± 7.8° (D_o/I_o = 9.7 ± 3.8°/35.1 ± 5.4°). The mean palaeolatitudes (with minimum and maximum latitudes) of the INC and SCB at the reference point in the Lower Cretaceous are calculated at 30.2°N (23.8°N and 37.8°N) and 19.4°N (15.9°N and 23.1°N), respectively (Table 3). There is thus a significant difference between the INC and SCB latitude during this time. This indicates a relative movement of the INC with respect to SCB.

From the Lower Cretaceous pole, a latitudinal translation of the INC of ca. −11.1 ± 6.2° or a southeastward displacement of about 1700 km along the Red River Fault and a clockwise rotation of 17.8 ± 6.8° with respect to the SCB is estimated. Carter and Bristow (2003) demonstrated that the palaeogeographic position of the Khorat Basin in Lower Cretaceous was probably close to the Sichuan Foreland Basin in the South China Block. In addition, geological evidences such as coinciding tectonic boundaries of western Sichuan and Khorat basins and matching of depositional/erosional boundaries of the Chuxiong and Khorat basins (Tapponnier et al., 1986; Peltzer and Tapponnier, 1988; Leloup et al., 1995). The evidence from Cretaceous granites in the western boundaries of the SCB and INC was observed as lateral offset along the ASRRF zone about 700 ± 100 km (Leloup et al., 1995). If it is reconstructed these locations, they could be close together at that time (Fig. 7b). The calculated Lower Cretaceous for the INC palaeolatitude of 30.2°N is in the same range as in the global tectonic reconstruction models (Seton et al., 2012; Metcalfe, 2013).

4.4.3. Upper Cretaceous time

The Upper Cretaceous palaeomagnetic pole derived from a combination of site means from the Bolikhamxay and Savannakhet areas is located at Plat./Plon. = 67.0°N/180.8°E, (A_{95} = 4.9°; Tables 2 and 3; Fig. 6). Even though there are signs of Middle-Upper Cretaceous tectonic activities in the Khorat Plateau (Lovatt Smith et al., 1996; Booth and Sattayarak, 2011) such as Phu Phan Uplift in Thailand, and Khammouane and Nam Leuk uplifts in Laos, these events seem to have not affected the ChRMs directions, possibility due to a low degree of internal deformation. The null test result of the mean directions and poles from the Khorat Plateau shows insignificant differences from the Lower Cretaceous. The result indicates that at least the central part of Indochina Block in this time was still tectonically intact before extrusion started after India–Asia collision.

The calculated directions of the INC (SCB) at the reference position from this pole are D_o/I_o = 24.8 ± 5.4°/43.3 ± 6.3° (D_o/I_o = 11.0 ± 3.4°/36.2 ± 4.7°). From the INC and SCB poles at the reference position, the palaeolatitudes (with minimum and maximum latitudes) are calculated at 25.2°N (20.6°N and 30.5°N) for Indochina, and at 20.1°N (17.0°N and 23.4°N) for the South China. This indicates that the reference location of the INC in the Upper Cretaceous was significantly different from the present position. This palaeoposition for Indochina is similar to the position (20–30°N) that Insamut et al. (1995) determined from palaeomagnetic data obtained from the Upper Cretaceous to Lower Tertiary redbeds of the Phu Tok Formation located northeast of the Khorat Basin, and Li et al. (2004) determined from rocks in the Khorat Plateau. Thus, the Indochina in the Upper Cretaceous time was probably located in the western part of South China (Fig. 7c). The poleward translation and rotation of the INC with respect to the SCB are estimated as −5.3 ± 4.7° or 750 km of lateral translation along the ASRR shear zone and clockwise rotation of 13.3 ± 5.0°, respectively (Table 3).

4.4.4. Southward displacement and rotation of the Indochina Block

Based on the palaeomagnetic data from the Lower Jurassic to Upper Cretaceous redbeds in the central part of Indochina, a lateral translation of the Indochina Block with respect to the South China since Lower Jurassic is indicated. The translation of ca. 750–1700 km occurred along the ASRR shear zone. Previous geological and palaeomagnetic interpretations reasonably well support the suggested southward displacement of the Indochina along the ASRRF. For example, a northward movement of the South China Block probably took place in Upper Jurassic along the ASRRF; whereas, the suggested ca. 950 ± 150 km movement of the Khorat Plateau in the Indochina Block was in a southeastward direction during Jurassic to Cretaceous times (Yang, 1992; Charusiri et al., 2006). Yang and Besse (1993) also suggested that the difference in latitude between Khorat and Sichuan Basin indicates a translation of Indochina Block relative to South China Block of ca. 11.5 ± 6.7° or 1500 ± 800 km along the ASRRF since the Cretaceous. A latitudinal translation of central Laos was estimated from palaeomagnetic data from Jurassic-Cretaceous redbeds (ca. 7.1 ± 2.2°; Takemoto et al., 2009). Lateral offsets of geologic markers from Mesozoic redbed basins across the ASRR shear zone, e.g. matching of boundaries of Chuxiong and Khorat basins, imply a southeastward displacement of ca. 1050 ± 100 km and a displacement of ca. about 800 ± 200 km is also indicated from similarities between the western Sichuan and Khorat basins (Leloup et al., 1995).

A pre-Tertiary reconstruction based on lateral offsets indicates a displacement of ca. 600–700 km along the ASRRF (Replumaz and Tapponnier, 2003; Replumaz et al., 2004; Faure et al., 2014). Ali et al. (2010) proposed a relative position of the INC versus SCB blocks, which suggests a ca. 500 km left-lateral displacement since the end of Middle Permian. Left-lateral offset along the ASRRF observed from Triassic volcanic arc at the northwest of Weixi, the SCB and the south of Dien Bien Phu, the INC ranges between 650 and 780 km (Leloup et al., 1995), and observed from matching Cretaceous granites at the west of Indochina and South China blocks yields a displacement of about 700 ± 100 km (Briais et al., 1993). U/Pb dating of metamorphic rocks in the Ailao Shan/Diancang Shan metamorphic belt along the Red River in Yunnan, South China indicated that there were significant movements of the blocks during Oligo-Miocene and that at least a 500 km southeastward displacement of the Indochina Block relative to South China occurred due to the India–Asia collision (Tapponnier et al., 1990). Displacement of the Indochina Block from recent GPS and geodetic observations the Indochina Block was estimated to have been moving toward the east at a rate of 8–10 mm/yr with respect to Eurasia (Calais et al., 2006; Vergnolle et al., 2007).

Our results indicate that the Indochina Block rotated ca. 13–18° clockwise with respect to the South China Block from Lower Jurassic to Upper Cretaceous times. Such a rotation of Indochina Block compared with South China is similar to what has been suggested in previous studies, e.g. clockwise rotation of ca. 14° (Yang and Besse, 1993), ca. 16–17° (Charusiri et al., 2006), and ca. 10° (Takemoto et al., 2009). Other studies from the northern part of Sundaland indicated a clockwise rotation (Yamashita et al., 2011); whereas rotation in the southern part was suggested to have been counter clockwise (Fuller et al., 1999; Hutchison, 2010; Yamashita et al., 2011). However, palaeomagnetic data from

Cretaceous rocks in Southern Vietnam, bounded by Pako and East Vietnam Boundary faults, did not indicate any rotation for southern part of Vietnam as compared to the Eurasia APWP (Otofuji et al., 2012; Chi and Dorobek, 2004). Otofuji et al. (2012) suggested that the lack of rotation indicated from palaeomagnetic data was probably due to local deformation of the southern part of the INC or due to a large counter clockwise rotation of the south of Sundaland. Palaeomagnetic studies results from the northwestern part of Vietnam (Takemoto et al., 2005) also indicated no significant rotation with respect to the Eurasia APWP. The calculation of translation and rotation from earlier palaeomagnetic data also indicated that there has been no rotation of the southern and northwestern Vietnam (Cung and Geissman, 2013).

4.4.5. Tectonic rigidity of the Indochina Block

Our palaeomagnetic result shows no significant difference of the ChRM directions or poles as compared with previous data from the central part of the Indochina Block. This may argue for the Khorat Plateau area and the central part of the Indochina acted as a rigid block since Lower Jurassic time. A P-wave model of the upper mantle beneath SE Asia at depth of 140–400 km revealed low P-wave velocity and continuality under the central part of the Indochina Block especially Khorat Plateau (Montelli et al., 2004; Li et al., 2006). This may suggest that the presence of continental roots beneath the Khorat Basin may be mechanically strong enough to resist the impact of tectonic deformation. However, this seismic anomalous zone shows clearly different from the northeastern part of Indochina Block (Red River Fault zone) and Shan Thai Block. This may support the internal and regional tectonic deformation of SE Asia. The relative translation and rotation of Indochina with respect to South China is most probably related to the extrusion tectonic model of Asia during Pre-Tertiary after the India–Asia collision.

5. Summary and conclusion

We demonstrate the positive fold tests for redbeds from the Lower and the Upper Cretaceous in Laos with exception for the Lower Jurassic redbeds from Champasak. The characteristic remanent magnetization of the Cretaceous redbeds is thus proven primary. Fold test for the Lower Jurassic redbeds is inconclusive but still not syn-fold or post-folding of remanences. This is probably due to a shallow bedding nature of these redbeds and due to a small contamination of a small fraction of secondary remanence carried by newly formed mineral, most probably magnetite at the time of folding or after. Palaeomagnetic poles calculated for Lower Jurassic, Lower Cretaceous and Upper Cretaceous redbeds are located at Plat./Plon.= 56.0°N/178.6°E (A_{95} = 2.6°), Plat./Plon. = 63.3°N/170.2°E (A_{95} = 6.9°) and Plat./Plon. = 67.0°N/180.8°E (A_{95} = 4.9°), respectively. On the basis of these results a translation of the Indochina Block about 5–11° or 750–1700 km along the Red River Fault and clockwise rotation of ca. 13–18° are estimated with respect to the South China Block. Our new palaeomagnetic results suggest that the central part of Indochina Block has been tectonically stable and acted as a rigid block since Lower Jurassic time. The estimated palaeoposition of the Khorat Plateau at ca. 21–26°N during Jurassic to Cretaceous argues for a close relation to the Sichuan Basin in the southwest of South China. These results support an earlier presented extrusion model for Indochina.

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